



UNDERGROUND CONCRETE PITS

PURPOSE

This document establishes guidelines for the design of underground concrete pits.

SCOPE

This document includes the following major sections:

- GENERAL
- DESIGN CONDITIONS
- BASIN DESIGN
- CONCRETE MIX REQUIREMENTS
- REINFORCING REQUIREMENTS
- PROTECTIVE LININGS
- JOINTS
- HYDRAULIC CONSIDERATIONS
- CATCH BASINS AND MANHOLES
- CIRCULAR CONCRETE TANKS
- SUPPLIER FURNISHED CONCRETE PITS
- REFERENCES
- ATTACHMENTS

APPLICATION

This document should be used for the design of underground concrete pits, sumps, boxes, or similar structures.

GENERAL

The most common types of underground pits encountered are sumps, manholes, valve and electrical pull boxes, and API (American Petroleum Institute) separators.

Sumps, manholes, and valve boxes vary in size, shape, and depth. They may or may not be covered with a concrete slab. Floors of these structures are usually sloped and may have a basin for cleanout purposes.

The typical API separator consists of a channel or channels of relatively long length and shallow depth to allow time for oil particles to surface and be removed by a slotted pipe. A series of vertical baffles near the channel inlet is used to distribute the flow uniformly. A sloping floor slab is used with basins located at the upstream end for the collection of sludge. A baffle, weir (used to separate clean water from oil), and sump pit are located at the downstream end.



UNDERGROUND CONCRETE PITS

DESIGN CONDITIONS

Load Cases

Consider the following loading conditions in the design of underground pits:

- Dead load - weight of pit
- Lateral earth pressure
- External water pressure
- Internal water pressure
- External surcharge - due to traffic or construction equipment
- Buoyancy

Internal hydrostatic pressure should be checked without soil pressure acting on outside of walls. External earth and surcharge pressure and buoyancy should be checked with pit empty.

Soil Bearing

Normally, soil bearing pressure is not excessive for concrete pits. It should be checked when the foundation is placed on weak soil or fill, or whenever a special type of foundation is used. If the structure rests on more than one kind of soil, necessary provisions should be made to prevent differential settlement.

Soils Information

Concrete pit walls should be designed for lateral earth pressure based on at rest or active state condition depending on the likelihood of the wall to move. The coefficient of earth pressure at rest (K_0) depends on the relative density of the soil and the process by which the deposit was formed. To change the coefficient of earth pressure to the active state, the soil must be free to expand laterally, which implies the top of the wall must move with the applied load. If the position of a wall is fixed, the earth pressure will likely be a value similar to the at rest earth pressure. Walls of pits analyzed using 2-way bending typically use at rest pressures, while cantilever walls are generally designed using active pressures. Lateral earth pressure coefficients can range from 0.3 for sands to 0.7 or more for clays and are largely affected by compaction methods used in backfilling. Typical earth pressure coefficients for various types of soil can be found in Bowles or similar soil mechanics textbooks. Appropriate pressure coefficients or equivalent fluid pressure diagrams should be obtained from the Geotechnical Engineer, as well as location of the ground water table, allowable soil bearing, sulfate content of soil and ground water, and recommended soil preparation.

Soil preparation under the base slab is necessary whenever weak or unstable soil conditions occur. Sand beds, gravel beds, stabilized material, or geotextiles have proven



UNDERGROUND CONCRETE PITS

successful in providing a stable surface on which to cast the base slab. A vapor barrier may be necessary when pouring slab over coarse sand or gravel to prevent loss of moisture into soil voids.

The possibility of creating a perched water table should be investigated when using granular backfill materials in excavations cut into clay or similar impervious soils.

BASIN DESIGN

Wall Design

Depending on the length-to-height relationship, the analysis of the walls should be performed as follows:

- If the length of the wall is more than 3 times its height, the wall is to be designed as a vertical cantilever. If the height of the wall is excessive, counterforted walls may be advantageous.
- If the length of the wall is equal to or less than 3 times its height, the wall is to be designed for 2-way bending using the coefficients in Engineering Monograph Number 27, Moments and Reactions for Rectangular Plates, by the United States Department of the Interior, or similar design aid such as Rectangular Concrete Tanks by the PCA (Portland Cement Association).

Reinforced concrete walls will be a minimum of 6 inches (150mm) thick with a single layer of reinforcing. Walls 10 inches (250mm) or thicker will have reinforcement on each face.

Minimum wall reinforcement, vertical and horizontal, will be according to ACI (American Concrete Institute) 318-08, Section 14.3, based on the wall cross section with continuous bars placed horizontally in the top of the wall and additional bars placed around the perimeter of openings.

Minimum bar size will be #4 (ϕ 12mm) and maximum spacing will be 12 inches (300mm).

The walls will be designed with contraction joints at 30ft (10meters) centers (maximum).

Joints will meet the requirements outlined in ACI 318-08, Section 6.4.

Slab Design

Top and bottom slabs will be designed as either 1-way or 2-way slabs. Strip analysis will be used for top slabs with numerous openings to determine the most critical section.



UNDERGROUND CONCRETE PITS

Moments transferred from wall into slab will be included in the design of slab reinforcing.

Minimum slab reinforcing will be that required for shrinkage and temperature stresses according to ACI 318-08, Section 7.12.

Reinforcing will be placed continuously across the entire slab. Whenever slab thickness is 10 inches (250mm) or more, 2 layers of reinforcing will be used.

Slab will be designed with contraction joints at 30ft (10meters) centers (maximum).

Shear keys and waterstops will be provided as required at the joint between the slab and the wall for watertightness. Gravel bottoms and French drains may also be utilized in areas of high water tables to maintain a dry pit.

Buoyancy

A minimum factor of safety against flotation of 1.2 will be used for the empty condition. Wall or slab thickness may be increased or an apron (toe) added to resist buoyant effects.

Make provisions during construction to prevent flotation of basins prior to backfilling. Flotation may be prevented by the use of openings or check valves.

CONCRETE MIX REQUIREMENTS

Concrete needed for most underground pit applications must be impervious. Low permeability and good durability are essential to prevent seepage and cracking, and to provide adequate protection to the reinforcing. Selection of cement type should be in accordance with ACI 318-08, Section 4.3. Water cement ratio (which is a function of f'_c) should be as low as possible, consistent with obtaining satisfactory workability, and of good densification. This water cement ratio will be in accordance with ACI 318-08, Chapter 4.

Concrete will be air entrained with a maximum slump of 3 inches (75mm).

The use of admixtures such as superplasticizers, water reducing agents, pozzolans, fly ash, silica fume, or other products available may be considered to achieve required imperviousness. Engineer will thoroughly investigate the effects that each of these admixtures has on concrete strength, slump, workability, durability, and overall cost. Trial mixes should be made with the admixture and job materials at temperatures and humidities anticipated on the job to determine the effects of the admixture on the properties of fresh and hardened concrete. Proper and complete curing is essential to minimize cracking.



UNDERGROUND CONCRETE PITS

Additional construction materials are covered in Structural Engineering Guideline 000.215.1239: Cooling Tower Foundations.

REINFORCING REQUIREMENTS

Provisions outlined in ACI 318-08, Section 10.5, are adequate to control cracks under most circumstances. Additional concrete cover may be regarded as sacrificial to corrosive exposure. Sanitary engineering structures and structures that must be absolutely watertight and resistant to chemicals should be designed according to the provisions outlined in ACI Committee 350, Concrete Sanitary Engineering Structures.

PROTECTIVE LININGS

Protective linings are often necessary in underground concrete pits to prevent deterioration of the concrete caused by the various corrosive products and byproducts produced in petrochemical plants. Research of available linings and their degree of protection, along with coordination between structural and process, is necessary to determine the best lining for a given commodity.

A wide variety of bitumastic compounds and epoxies are available as liners or coatings to enhance leak prevention and prevent concrete deterioration. Manufacturer's recommendations and application procedures should be obtained for each specific use.

Severe corrosive environments and high temperatures may warrant the use of acid brick lining, fire brick lining, or double wall pits.

JOINTS

Types of joints commonly used are addressed in Structural Engineering Guideline 000.215.1239: Cooling Tower Foundations.

HYDRAULIC CONSIDERATIONS

Hydraulic factors unique to the particular function of the underground concrete pit should be considered in the layout and design of the pit, which is usually a joint effort of the Process, Mechanical, and Structural groups. There are definite hydraulic factors to consider in the design of a sump in order to prevent turbulence in the pump sump, and thus, prolong the life of the pumps. The hydraulic factors given in the procedure have been compiled from material by the Peerless Pump Company and Standards of the Hydraulic Institute. (Refer to Structural Engineering Guideline 000.215.1239.)

The ideal flow to a pump is a straight channel with no obstructions or changes in direction. Water should not flow past one pump to reach another.



UNDERGROUND CONCRETE PITS

The bottom of the sump should be flat and level, in the direction of flow, for a minimum of 5-1/2 D (D = diameter of pump suction bell) from the centerline of the pump.

Suggested sump dimensions for proper pump operation are given in Structural Engineering Guideline 000.215.1239. The graph should be entered with the rated capacity of the pump to obtain the required dimensions.

Maximum flow in sump will not exceed one foot per second (0.3m/s).

CATCH BASINS AND MANHOLES

Catch basins and manholes are typically constructed from precast sections of reinforced concrete pipe with cast in place or precast bottom slabs. Top lids also may be cast in place or precast and contain a cast iron manhole ring and cover. Covers may be open grate, solid, or gasket sealed and designed for light or heavy-duty depending on traffic loading. Manufacturer's literature should be consulted for exact dimensions and details.

When using precast sections for manholes and catch basins, the possibility of frost or freeze action lifting the upper sections should be investigated. Hold-down straps or similar devices should be used in areas where this possibility exists.

CIRCULAR CONCRETE TANKS

Procedures and suggested details for the design of circular concrete tanks can be found in Circular Concrete Tanks without Prestressing by PCA.

SUPPLIER FURNISHED CONCRETE PITS

Pre-manufactured double wall pits as well as a variety of precast manholes, sumps, valve boxes, lids, and catch basins are available and may aid constructibility.

REFERENCES

ACI (American Concrete Institute) 318-08

ACI (American Concrete Institute) Committee 350, Concrete Sanitary Engineering Structures

Moments and Reactions for Rectangular Plates. Engineering Monograph Number 27. United States Department of the Interior. Bureau of Reclamation. 1990.

PCA (Portland Cement Association) Publications



UNDERGROUND CONCRETE PITS

IS072-1992 Circular Concrete Tanks without Prestressing

S001-2007 Effects of Substances on Concrete and Guide to Protective Treatment

IS003-1998 Rectangular Concrete Tanks

Structural Engineering:

[Guideline 000.215.1239](#)

Cooling Tower Foundations

ATTACHMENTS

Attachment 01: Sample Design 1: Underground Valve Box

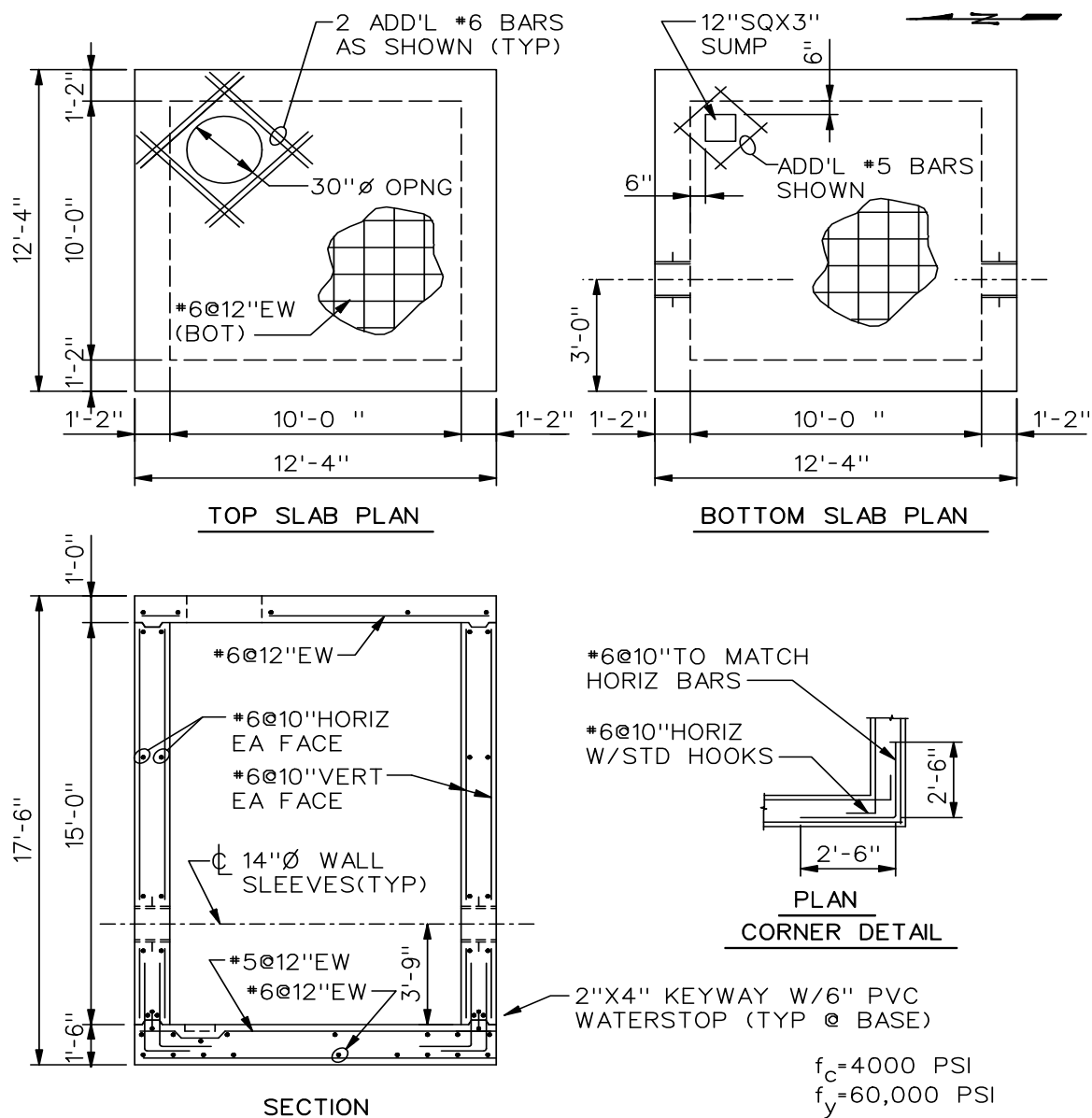
Attachment 02: Sample Design 2: API Separator

Note: *Calculations for Attachment 01 and 02 have not been updated to meet the requirements of the latest references and design codes.*

UNDERGROUND CONCRETE PITS

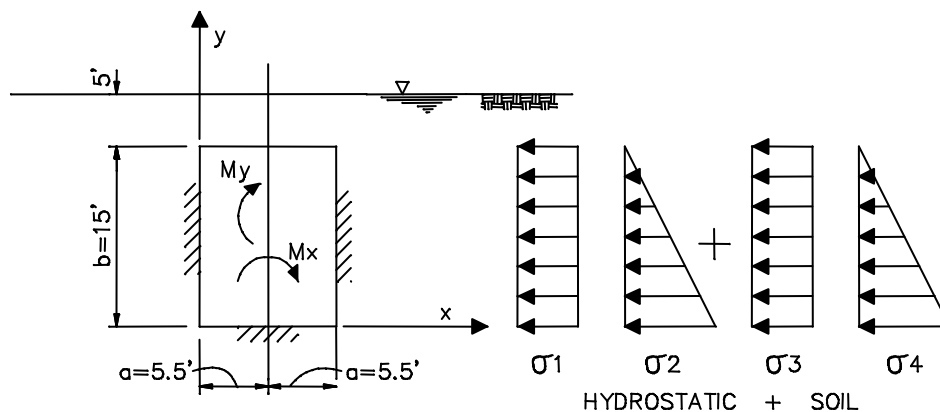
Sample Design 1: Underground Valve Box

Note: Calculations for Attachment 01 have not been updated to meet the requirements of the latest references and design codes.



UNDERGROUND CONCRETE PITS

Sample Design 1: Underground Valve Box



Design Criteria and Codes:

Reference: ACI 318-05 and Engineering Monograph No. 27 - United States Department of Interior
Bureau of Reclamation, 1990.

Concrete: $f'_c = 4000$ PSI Reinforcing: $f_y = 60000$ PSI

Soil Conditions: Sandy Clay (assumed) Assume water table at ground surface

Allowable soil bearing: 3.0 KSF

Soil weight: $\gamma = 120$ PCF

Submerged soil weight: 68 PCF

Coeff. of lateral earth pressure: $K_o = 0.7$ (at rest)

Sump has 10'-0" x 10'-0" inside dimensions
Plate fixed on three sides; hinged at the top

Lateral Loads:

$$\sigma_1 = 0.0624 (5.0) = 0.31 \text{ KSF}$$

$$\sigma_2 = 0.0624 (20.0) - 0.31 = 0.94 \text{ KSF}$$

$$\sigma_3 = K_o \gamma_{\text{sub}} h = 0.70 (0.068) (5.0) = 0.24 \text{ KSF}$$

$$\sigma_4 = 0.70 (0.068) (20.0) - 0.24 = 0.71 \text{ KSF}$$

Hydrostatic pressure

Hydrostatic pressure

Soil pressure

Soil pressure



UNDERGROUND CONCRETE PITS

Sample Design 1: Underground Valve Box

Solution:

Find M_x : (Reference Monograph No. 27)

$a/b = 5.5 / 15 = 0.37$ Therefore, USE 3/8 Column of table

PG	LOAD	x/a = 0	y/b = .6	x/a = 0	y/b = .4	x/a = 1.0	y/b = .6	x/a = 1.0	y/b = .4
16	$\sigma_1 + \sigma_3 = 0.55$	$.0403(.55)(15)^2 = 4.99$	$.0374(.55)(15)^2 = 4.63$	$-.0197(.55)(15)^2 = -2.44$	$-.0181(.55)(15)^2 = -2.24$				
19	$\sigma_2 + \sigma_4 = 1.65$	$.017(1.65)(15)^2 = 6.31$	$.0207(1.65)(15)^2 = 7.68$	$-.0084(1.65)(15)^2 = -3.12$	$-.0098(1.65)(15)^2 = -3.64$				
		11.30 K-FT	$M_x = 12.31 \text{ K-FT}$		-5.56 K-FT				-5.88 K-FT

Maximum for horizontal

Find M_u : $a/b = 3/8$

$x/a = 1.0$; $y/b = 0$ is critical location

$$\sigma_1 + \sigma_3 = 0.55 \quad M_y = .0288(.55)(15)^2 = 3.56 \text{ K-FT}$$

$$\sigma_2 + \sigma_4 = 1.65 \quad M_y = .0200(1.65)(15)^2 = \underline{7.43 \text{ K-FT}}$$

$$\text{Total } M_y = \underline{10.99 \text{ K-FT}} \quad (\text{Maximum for vertical})$$

Since moments are close to being the same, design both horizontal & vertical reinforcing for the larger of the two.

$$M_u = 1.4 (12.31) = 17.23 \text{ K-FT} \quad (\text{Since hydro \& soil pressures are similar, USE avg factor } 1/2(1.2+1.6) = 1.4)$$

$$\text{Assume wall thickness} = 14" \quad d = 14" - 2" - 1/2(1") = 11.5"$$

$$F = \frac{bd^2}{12000} = \frac{12(11.5)^2}{12000} = 0.132$$

$$K_u = \frac{M_u}{F} = \frac{17.23}{0.132} = 130.53 \quad a_u = 4.39$$



UNDERGROUND CONCRETE PITS

Sample Design 1: Underground Valve Box

$$A_s = \frac{M_u}{a_u d} = \frac{17.23}{4.39(11.5)} = 0.34 \text{ in}^2/\text{FT}$$

$$\frac{4}{3} A_s = \frac{4}{3} (0.34) = 0.45 \text{ in}^2/\text{FT}$$

$$A_{s\min} = 0.0033(12)(11.5) = 0.46 \text{ in}^2/\text{FT}$$

USE: #6 @ 10" E.W. ($A_s = 0.53 \text{ in}^2/\text{FT}$)

Check shear:

Condition for maximum shear: R_y critical over R_x

For $a/b = 3/8$ @ $x/a = 1.0$

$$R_y = 0.3427(0.55)(15) + 0.2647(1.65)(15) = 9.38 \text{ K / FT}$$

$$(\sigma_1 + \sigma_3) (\sigma_2 + \sigma_4)$$

$$V_u = 1.4 (9.38) = 13.13 \text{ K}$$

$$v_u = \frac{13.13(1000)}{0.85(12)(11.5)} = 111.9 \text{ PSI} \leq 2\sqrt{f'_c} = 126.5 \text{ PSI} \quad \text{OK}$$

USE: 14" Thick walls

Top slab design: (Assume 12" minimum thickness)

For simplicity, assume 1 FT strip simple span section

$$w = 1 \text{ FT} (0.15) + 0.0624(5 \text{ FT}) + 0.068(5 \text{ FT}) = 0.80 \text{ K / FT}$$

$$V_{\max} = w/2 = 0.80(10) / 2 = 4.0 \text{ K / FT}$$

$$V_u = 1.4 (4.0) = 5.6 \text{ K / FT}$$

UNDERGROUND CONCRETE PITS

Sample Design 1: Underground Valve Box

$$M_{\max} = wl^2/8 = 0.80(10)^2 / 8 = 10.0 \text{ K' / FT}$$

$$M_u = 1.4(10.0) = 14.0 \text{ K' / FT}$$

$$d_{\min} = 12'' - 2'' - 1/2(1'') = 9.5''$$

$$v_u = \frac{5.6(1000)}{0.85(12)(9.5)} = 57.8 \text{ PSI} \leq 2\sqrt{f'_c} \quad \text{OK}$$

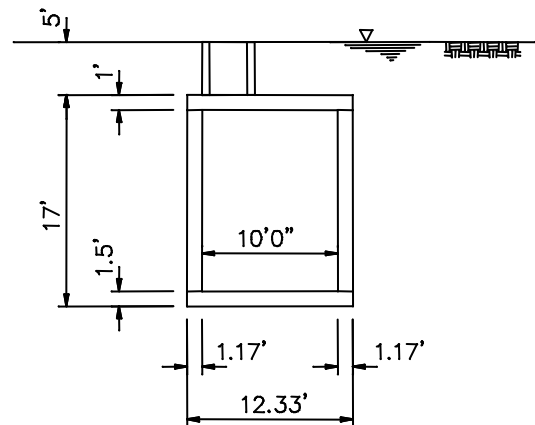
$$F = \frac{bd^2}{12000} = \frac{12(9.5)^2}{12000} = 0.090$$

$$K_u = \frac{M_u}{F} = \frac{14.0}{0.090} = 155.6 \quad a_u = 4.37$$

$$A_s = \frac{M_u}{a_u d} = \frac{14.0}{4.37(9.5)} = 0.34 \text{ in}^2 / \text{FT}$$

$$\frac{4}{3} A_s = \frac{4}{3} (0.34) = 0.45 \text{ in}^2 / \text{FT}$$

$$A_{s\min} = 0.0033(12)(9.5) = 0.38 \text{ in}^2 / \text{FT}$$



USE: #6 @ 10" E.W. (Bottom) $(A_s = 0.52 \text{ in}^2 / \text{FT})$

Check Buoyancy: (Empty case)

Assume water table @ ground surface

Weights:

Bottom slab = $(12.33)^2(1.5)(0.15)$	=	34.2 K
Walls = $4(15 \times 10)(1.17)(0.15)$	=	105.3 K
Top slab = $(12.33)^2(1.0)(0.15)$	=	22.8 K
Soil (saturated) = $(12.33)^2(5.0)(0.068)$	=	<u>51.7 K</u>
Total Wt.	=	214.0 K

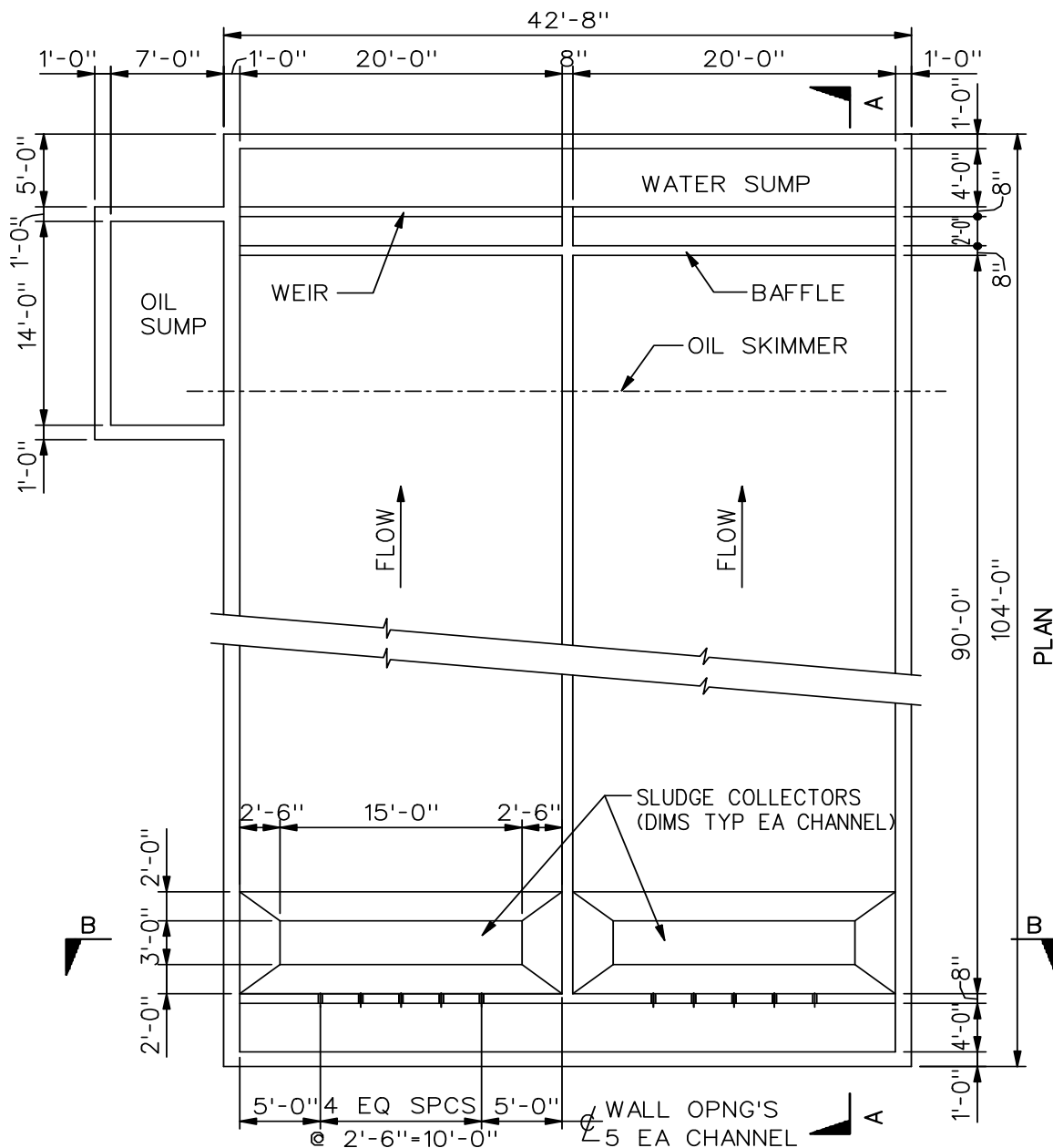
$$\text{Buoyant force} = (12.33)^2 (17.0) (0.0624) = \mathbf{161.3 \text{ K}}$$



UNDERGROUND CONCRETE PITS

Sample Design 2: API Separator

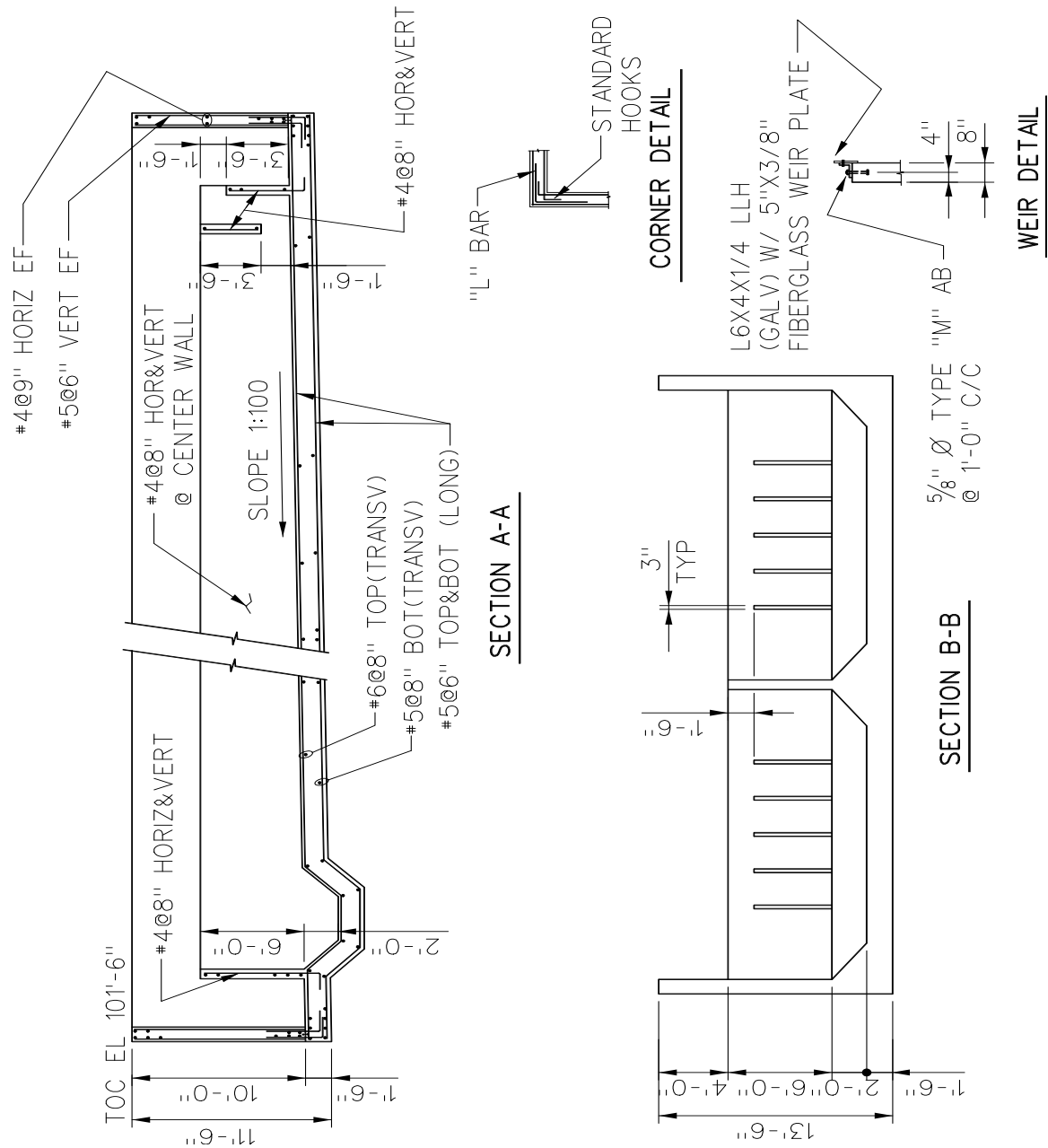
Note: Calculations for Attachment 02 have not been updated to meet the requirements of the latest references and design codes.





UNDERGROUND CONCRETE PITS

Sample Design 2: API Separator





UNDERGROUND CONCRETE PITS

Sample Design 2: API Separator

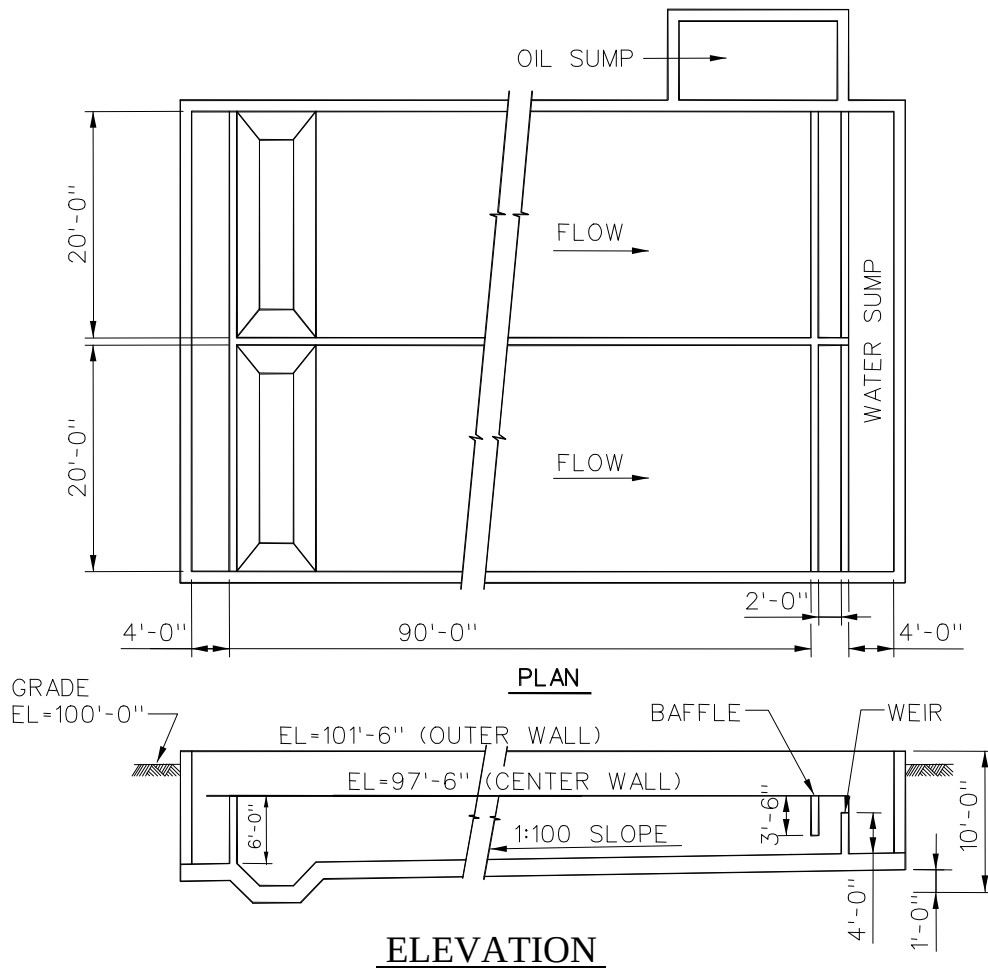
DESIGN CRITERIA AND CODES

ACI 318-89 USD

CONCRETE: $f'_c = 4000$ PSI

REINFORCING: $f_y = 60,000$ PSI

SOIL PARAMETERS: ALLOW SOIL BEARING = 3 KSF
ANGLE OF INTERNAL FRICTION $\phi = 15^\circ$
SOIL WEIGHT $\gamma = 110$ PCF
BUOYANT SOIL WEIGHT $\gamma = 70$ PCF
GROUND WATER LEVEL = ELEV 95'-0"





UNDERGROUND CONCRETE PITS

Sample Design 2: API Separator

Exterior wall design:

Due to the large length to height ratio, side walls will be designed as vertical cantilever.

CASE I: (Basin full of water) (USE: 1 FT strip)

$$P_w = 0.0624 (10')^2 (1/2) = 3.12 \text{ K}$$

$$M = 3.12 \text{ K} (10') (1/3) = 10.4 \text{ K' / FT}$$

CASE II: (Basin empty - External earth pressure)

$$\text{Coeff. of active earth pressure: } K_a = \tan^2 \left(45 - \frac{\phi}{2} \right)$$

$$K_a = \tan^2 \left(45 - \frac{15}{2} \right) = 0.59$$

$$w_1 = 0.110 (5.0') (0.59) = 0.32 \text{ KSF}$$

$$w_2 = 0.32 + 3.5' ((0.070) (0.59) + 0.0624) = 0.68 \text{ KSF}$$

$$P = P_1 + P_2 + P_3 = 1/2 (0.32) (5.0') + 0.32(3.5') + 1/2 (3.5') (0.68 - 0.32)$$

$$P = 0.80 + 1.12 + 0.63 = 2.55 \text{ K / FT}$$

$$M = 0.80 (3.5 + 5.0 / 3) + 1.12 (3.5 / 2) + 0.63 (3.5 / 3) = 6.8 \text{ K' / FT}$$

Reinforcing: Case I controls (Use same reinforcement each face)

$$V_u = 1.4 (3.12) = 4.37 \text{ K / FT} \quad M_u = 1.4 (10.4) = 14.6 \text{ K' / FT}$$

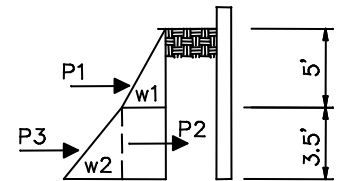
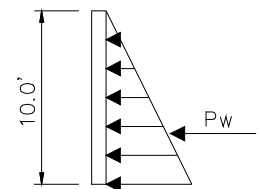
$$\text{Assume 12" wall} \quad d = 12" - 2" - 1/2(1") = 9.5"$$

$$v_u = \frac{V_u}{\phi b d} = \frac{4.37}{0.85(12)(9.5)} = 45 \text{ PSI} \leq 2\sqrt{f'_c} = 126.5 \text{ PSI} \quad \text{O.K.}$$

$$F = \frac{b d^2}{12000} = \frac{12(9.5)^2}{12000} = 0.090$$

$$K_u = \frac{M_u}{F} = \frac{14.6}{0.090} = 162 \quad a_u = 4.38$$

$$A_s = \frac{M_u}{a_u d} = \frac{14.6}{4.38(9.5)} = 0.35 \text{ in}^2/\text{FT}$$





UNDERGROUND CONCRETE PITS

Sample Design 2: API Separator

$$\frac{4}{3} A_s = \frac{4}{3} (0.35) = 0.47 \text{ in}^2/\text{FT}$$

$$A_{s\min} = 0.0033(12)(9.5) = 0.38 \text{ in}^2/\text{FT}$$

USE: #5 @ 9" each face (vertical)	($A_c = 0.41 \text{ in}^2/\text{FT}$)
-----------------------------------	---

Horizontal steel: (Per ACI 318-05, Section 14.3)

$$\text{Min. steel} = 0.0020(12)(12) = 0.36 \text{ in}^2 = 0.15 \text{ in}^2 / \text{face}$$

USE: #4 @ 12" each face (horizontal)	($A_c = 0.20 \text{ in}^2 / \text{FT}$)
--------------------------------------	---

Check crack control: (Per ACI 318-05, Section 10.6)

$$s = 15(40000/f_s) - 2.5 C_c < 12(40000/f_s) \quad f_s = 0.667(60000 \text{ PSI}) = 40000 \text{ PSI}$$

$$C_c = 2.3"$$

$$s = 15(40000/40000) - 2.5(2.3) = 9.25 \text{ in} < 12(40000/40000) = 12 \text{ in}$$

USE: 8" spacing to satisfy crack control
--

Center wall design: (Assume 8" wall w/ single layer reinforcement)

Design as cantilever with water in one channel (1'-0" strip)

$$P_w = 0.0624(6')^2 (1/2) = 1.12 \text{ K}$$

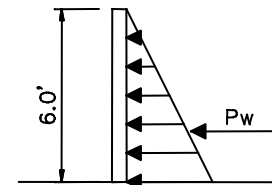
$$V_u = 1.4(1.12) = 1.57 \text{ K / FT} \quad M_u = 1.57(6') 1/3 = 3.14 \text{ K' / FT}$$

$$v_u = \frac{1.57(1000)}{0.85(12)(4)} = 38 \text{ PSI} \leq 2\sqrt{f'_c} = 126.5 \text{ PSI} \quad \text{O.K.}$$

$$F = \frac{bd^2}{12000} = \frac{12(4)^2}{12000} = 0.016$$

$$K_u = \frac{M_u}{F} = \frac{3.14}{0.016} = 196 \quad a_u = 4.35$$

$$A_s = \frac{M_u}{a_u d} = \frac{3.14}{4.35(4)} = 0.18 \text{ in}^2/\text{FT}$$





UNDERGROUND CONCRETE PITS

Sample Design 2: API Separator

$$\frac{4}{3} A_s = \frac{4}{3} (0.18) = 0.24 \text{ in}^2/\text{FT}$$

$$A_{s\min} = 0.0033(12)(4) = 0.16 \text{ in}^2/\text{FT}$$

USE: #4 @ 10" single layer (vertical) ($A_s = 0.24 \text{ in}^2/\text{FT}$)

Horizontal Steel: (Per ACI 318-05, Section 14.3)

$$\text{Min. steel} = 0.0025 (12) (8) = 0.24 \text{ in}^2$$

USE: #4 @ 10" single layer (horizontal) ($A_s = 0.24 \text{ in}^2/\text{FT}$)

Check crack control: (Per ACI 318-05, Section 10.6)

$$s = 15 (40000/f_s) - 2.5 C_c < 12 (40000/f_s)$$

$$f_s = 0.667(60000 \text{ PSI}) = 40000 \text{ PSI}$$

$$C_c = 4"; \text{ use } 2" \text{ max.}$$

$$s = 15(40000/40000) - 2.5 (2.0) = 9.00 \text{ in} < 12 (40000/40000) = 12 \text{ in}$$

USE: 8" spacing to satisfy crack control

Interior wall design:

Interior walls and weirs designed in a similar manner

USE: 8" thick walls w/ #4 @ 10" single layer (vertical & horizontal)

Baffle design:

USE: Same wall thickness and reinforcing as interior walls

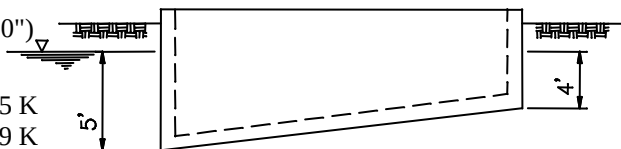
Check Buoyancy: (High ground water elev. = 95'-0")

Weight of structure:

$$\text{Slab} = 104'(42.67')(1.5)(0.15) = 998.5 \text{ K}$$

$$\text{Endwalls} = (9'+10')(1')(40.67')(0.15) = 115.9 \text{ K}$$

$$\text{Sidewalls} = 2(104')(9'+10')(1/2)(1')(0.15) = 296.4 \text{ K}$$





UNDERGROUND CONCRETE PITS

Sample Design 2: API Separator

Center wall = $92'(5'+6')(1/2)(8/12)(0.15)$	=	50.6 K
Interior walls = $40.67'(6')(8/12)(0.15)$	=	24.4 K
Weir = $4'(40.67')(8/12)(0.15)$	=	16.3 K
Baffle = $3.5'(40.67')(8/12)(0.15)$	=	<u>14.2 K</u>
Total	=	1516.3 K

Upward buoyant force = $(4'+5') (1/2) (104') (42.67') (0.0624) = 1246.1k$

FS = $1516.3 K / 1246.1 K = 1.22 > 1.20$ **O.K.**

Base slab design:

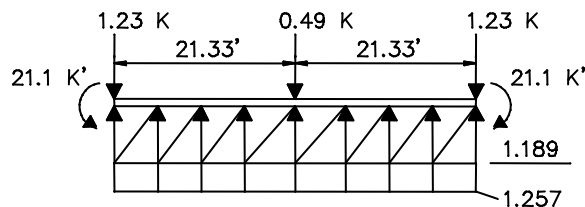
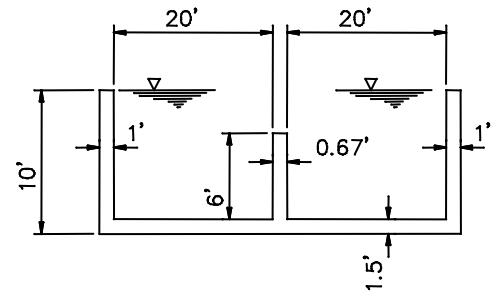
CASE I: (Basin full of water)

Weight per foot:

Slab = $1.5'(42.67') (0.15)$	=	9.6 K
End walls = $2(1') (10') (0.15)$	=	3.0 K
Center wall = $0.67'(6') (0.15)$	=	0.6 K
Water = $(40.67'(10') - 6'(0.67')) (0.0624)$	=	<u>25.1 K</u>
Total	=	38.3 K

S.B. = $38.3 K / 42.67' = 0.898 \text{ KSF} < \text{Allowable}$ **O.K.**

Ult. S.B. = $1.4 (0.898) = 1.257 \text{ KSF}$



$M_u @ \text{slab base} = 1.4 (0.0624) (10')^2 (1/2) (10/3 + 1.5') = 21.1 \text{ K-FT}$

Water & Slab (Ult) = $1.4 [1.5(0.15) + 10'(0.0624)] = 1.189 \text{ KSF}$

Ult. net load @ end walls = $10'(1') (0.15 - 0.0624) (1.4) = 1.23 \text{ K}$

Ult. net load @ center wall = $6'(0.67') (0.15 - 0.0624) (1.4) = 0.49 \text{ K}$

Maximum moment occurs at centerline

$M_{u \max} (-) = 21.1 \text{ K-FT} + 1.23 \text{ K} (21.33') - (1.257 - 1.189) (21.33')^2 (1/2) = 31.9 \text{ K-FT (Top reinforcing)}$



UNDERGROUND CONCRETE PITS

Sample Design 2: API Separator

CASE II (Basin empty w/ external earth pressure)

$$w_1 = 0.32 \text{ KSF}$$

$$w_2 = 0.32 + 5.0' ((0.070) (0.59) + 0.0624) = 0.84 \text{ KSF}$$

$$P = P_1 + P_2 + P_3 = 1/2 (0.32) (5.0') + 0.32(5.0') + 1/2 (5.0') (0.84 - 0.32)$$

$$P = 0.80 + 1.60 + 1.30 = 3.70 \text{ K / FT}$$

$$M_u \text{ @ slab base} = 1.4 [0.8(5' + 5'/3) + 1.6(5'/2) + 1.3(5'/3)] = 16.1 \text{ K-FT}$$

Weight per foot:

Slab	=	9.6 K
End walls	=	3.0 K
Center wall	=	<u>0.6 K</u>
Total	=	13.2 K

$$\text{S.B.} = 13.2 \text{ K} / 42.67' = 0.309 \text{ KSF} < \text{Allowable} \quad \text{OK}$$

$$\text{Ult. S.B.} = 1.4 (0.309) = 0.433 \text{ KSF}$$

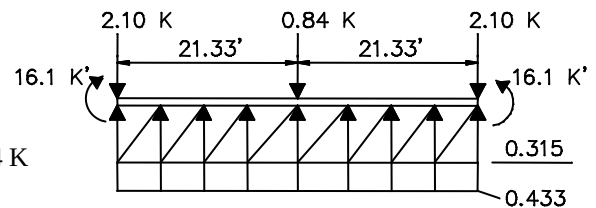
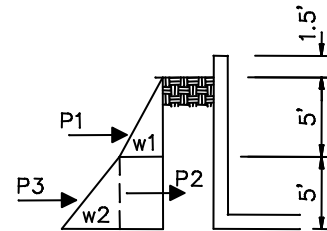
$$\text{Ult. slab weight} = 1.4 (1.5) (0.15) = 0.315 \text{ KSF}$$

$$\text{Ult. net load @ end walls} = 10'(1') (0.15) (1.4) = 2.10 \text{ K}$$

$$\text{Ult. net load @ center wall} = 6'(0.67') (0.15) (1.4) = 0.84 \text{ K}$$

Maximum moment occurs at end

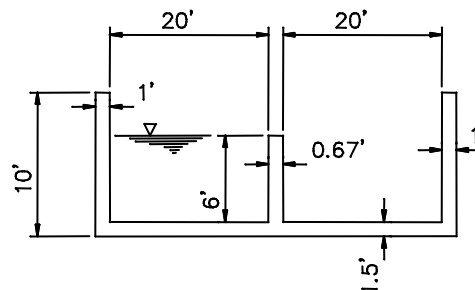
$$M_{u\max(+)} = 16.1 \text{ K-FT} \quad (\text{Bottom reinforcing})$$



CASE III (One channel of basin full)

Weight per foot:

Slab	=	9.6 K
End walls	=	3.0 K
Center wall	=	0.6 K
Water = 20'(6') (0.0624)	=	<u>7.49 K</u>
Total	=	20.69 K





UNDERGROUND CONCRETE PITS

Sample Design 2: API Separator

$$M = 10.33' P_w = 10.33' (7.49 \text{ K}) = 77.4 \text{ K-FT}$$

$$e = M / P = 77.4 \text{ K-FT} / 20.69 \text{ K} = 3.74'$$

$$\text{ULT S.B.} = \frac{P}{A} \left(1 \pm \frac{6e}{B} \right) = \frac{1.4(20.69)}{42.67} \left(1 \pm \frac{6(3.74')}{42.67} \right) = 1.036 \text{ KSF}$$

$$= 0.322 \text{ KSF}$$

$$\text{Wt. water \& slab (Ult.)} = 1.4[1.5(0.15) + 6'(0.0624)] = 0.839 \text{ KSF}$$

$$\text{Ult. slab weight} = 1.4 (1.5) (0.15) = 0.315 \text{ KSF}$$

$$\text{Ult. net load left wall} = 1.4 [4'(0.15) + 6'(0.15 - 0.0624)] = 1.58 \text{ K}$$

$$\text{Ult. net load center wall} = 1.4(0.15 - 0.0624) (6') (0.67') = 0.49 \text{ K}$$

$$\text{Ult. net load right wall} = 1.4(10.0') (0.15) = 2.10 \text{ K}$$

$$M_u @ \text{ slab base} = 1.4 (0.0624) (6')^2 (1/2) (6/3 + 1.5') = 5.50 \text{ K-FT}$$

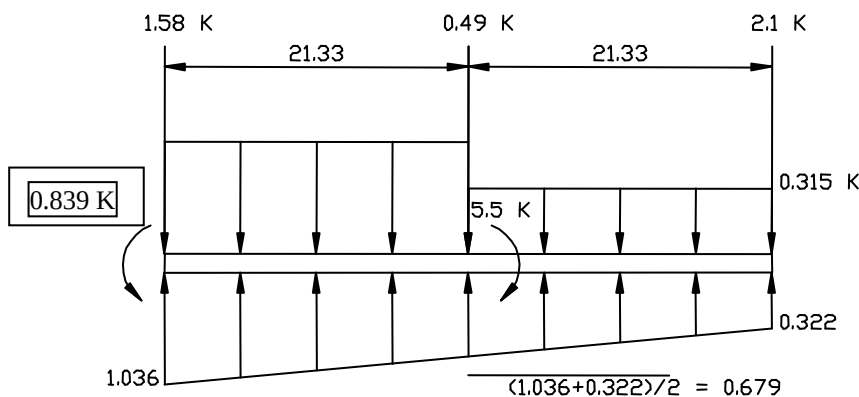
$$M_u = 2.1x - (0.322 - 0.315) \left(\frac{x^2}{2} \right) - \frac{1}{2} (x)(0.679 - 0.322) \left(\frac{x}{21.33} \right) \left(\frac{x}{3} \right)$$

$$M_u = 2.1x - 0.0035x^2 - 0.0028x^3$$

$$\frac{dM_u}{dx} = 0 = 2.1 - 0.007x - 0.0084x^2$$

$$x = \frac{0.007 \pm \sqrt{(0.007)^2 - 4(2.1)(-0.0084)}}{2(-0.0084)} = 15.42'$$

$$M_u = 2.1(15.42) - 0.0035 (15.42)^2 - 0.0028 (15.42)^3 = 21.3 \text{ K-FT (Top reinforcing)}$$





UNDERGROUND CONCRETE PITS

Sample Design 2: API Separator

Controlling slab design moments:

Top reinforcing: $M_u = 31.9 \text{ K-FT}$

Bottom reinforcing $M_u = 16.1 \text{ K-FT}$

For 18" thick slab: $d_{\text{top}} = 18" - 2" - 1/2(1") = 15.5"$

$$d_{\text{bottom}} = 18" - 3" - 1/2(1") = 14.5"$$

Top reinforcing:

$$F = \frac{bd^2}{12000} = \frac{12(15.5)^2}{12000} = 0.24$$

$$K_u = \frac{M_u}{F} = \frac{31.9}{0.24} = 133 \quad a_u = 4.39$$

$$A_s = \frac{M_u}{a_u d} = \frac{31.9}{4.39(15.5)} = 0.47 \text{ in}^2/\text{FT}$$

$$\frac{4}{3}A_s = \frac{4}{3}(0.47) = 0.63 \text{ in}^2/\text{FT}$$

$$A_{s\text{min}} = 0.0033(12)(15.5) = 0.62 \text{ in}^2/\text{FT}$$

USE: #6 @ 8" o.c. Transverse (Top) $(A_s = 0.66 \text{ in}^2/\text{FT})$
--

Bottom reinforcing:

$$F = \frac{bd^2}{12000} = \frac{12(14.5)^2}{12000} = 0.21$$

$$K_u = \frac{M_u}{F} = \frac{16.1}{0.21} = 77 \quad a_u = 4.44$$

$$A_s = \frac{M_u}{a_u d} = \frac{16.1}{4.44(14.5)} = 0.25 \text{ in}^2/\text{FT}$$

$$\frac{4}{3}A_s = \frac{4}{3}(0.25) = 0.33 \text{ in}^2/\text{FT}$$



UNDERGROUND CONCRETE PITS

Sample Design 2: API Separator

$$A_{\text{min}} = 0.0033(12)(14.5) = 0.57 \text{ in}^2/\text{FT}$$

USE: #5 @ 8" o.c. Transverse (Bottom) ($A_s = 0.46 \text{ in}^2/\text{FT}$)

Longitudinal steel:

(Temperature & shrinkage as a minimum)

$$A_s = 0.0018(18)(12) = 0.39 \text{ in}^2/\text{FT} = 0.20 \text{ in}^2/\text{FT} \text{ (Top \& Bottom)}$$

OR steel required to take wall moment

USE: #4 @ 10" o.c. Longitudinal (Top & Bottom) ($A_s = 0.24 \text{ in}^2/\text{FT}$) min.

USE: #5 @ 6" o.c. spacing to match vertical wall reinforcement



UNDERGROUND CONCRETE PITS

Sample Design 1: Underground Valve Box

F.S. against floatation = $214 / 161.3 = 1.32 > 1.2$ OK

Check soil bearing: (Assume operating case - full of water)

Structure & soil	=	214.0 K
Water = $(10.0)^2(14.5)(0.0624)$	=	<u>90.5 K</u>
Total	=	304.5 K

$$S.B. = \frac{304.5}{(12.33)(12.33)} = 2.00 \text{ KSF} \leq 3.00 \text{ KSF} \quad \text{OK}$$

Bottom Slab Design (Assume 1'-6" thick)

Bottom reinforcing: Design to carry the same vertical cantilever moment at the bottom of the wall.

$$M_y = 10.99 \text{ K-FT} \quad M_u = 1.4(10.99) = 15.39 \text{ K-FT}$$

$$d = 18" - 3" - 1/2(1") = 14.5"$$

$$F = \frac{bd^2}{12000} = \frac{12(14.5)^2}{12000} = 0.210$$

$$K_u = \frac{M_u}{F} = \frac{15.39}{0.210} = 73.3 \quad a_u = 4.44$$

$$A_s = \frac{M_u}{a_u d} = \frac{15.39}{4.44(14.5')} = 0.24 \text{ in}^2/\text{FT}$$

$$\frac{4}{3} A_s = \frac{4}{3}(0.24) = 0.32 \text{ in}^2/\text{FT}$$

$$A_{smin} = 0.0033(12)(14.5) = 0.57 \text{ in}^2/\text{FT}$$

USE: #6 @ 12" E.W. (Bottom) $(A_s = 0.44 \text{ in}^2/\text{FT})$	caused by soil bearing (empty case)
---	-------------------------------------

$$w = 214.0 \text{ K} / (12.33')^2 = 1.41 \text{ KSF} \quad w_{net} = (1.41 - 1.5(0.150)) = 1.19 \text{ KSF}$$



UNDERGROUND CONCRETE PITS

Sample Design 1: Underground Valve Box

Assume plate fixed along four edges with applied uniform load

Reference: Monograph No. 27 pg. 40 $a = b = 12.33 \text{ FT}$ $a / b = 1.0$

$x/a = 0$; $y/b = 0.5$ Critical location

$$M_x = 0.0500(1.19)(12.33)^2 = 9.05 \text{ K-FT} \quad M_u = 1.4(9.05) = 12.7 \text{ K-FT}$$

$$R_x = 0.4389(1.19)(12.33) = 6.44 \text{ K} \quad V_u = 1.4(6.44 \text{ K}) = 9.1 \text{ K}$$

$$d = 18" - 2" - 1/2(1") = 15.5"$$

$$v_u = \frac{9.1(1000)}{0.85(12)(15.5)} = 57.6 \text{ PSI} \leq 2\sqrt{f'_c} \quad \text{OK}$$

$$F = \frac{bd^2}{12000} = \frac{12(15.5)^2}{12000} = 0.240$$

$$K_u = \frac{M_u}{F} = \frac{12.7}{0.240} = 52.9 \quad a_u = 4.45$$

$$A_s = \frac{M_u}{a_u d} = \frac{12.7}{4.45(15.5)} = 0.18 \text{ in}^2/\text{FT}$$

$$\frac{4}{3}A_s = \frac{4}{3}(0.18) = 0.24 \text{ in}^2/\text{FT}$$

$$A_{smin} = 0.0033(12)(15.5) = 0.61 \text{ in}^2/\text{FT}$$

USE: #5 @ 12" E.W. (Top) ($A_s = 0.31 \text{ in}^2/\text{FT}$)
